



National Oceanic and
Atmospheric Administration
U.S. Department of Commerce

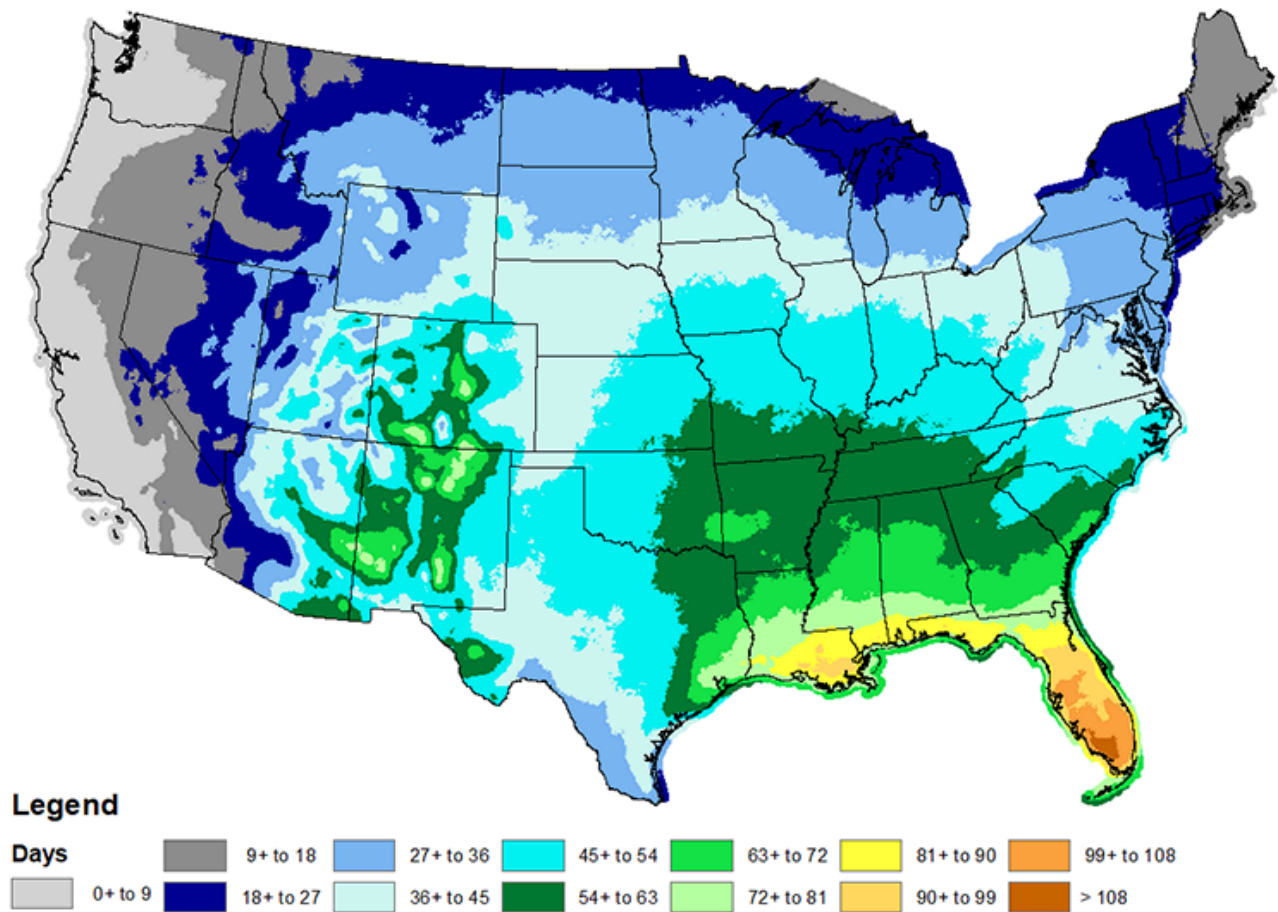
Thunderstorms

Introduction to Thunderstorms

It is estimated that there are as many as 40,000 thunderstorm occurrences each day world-wide. This translates into an astounding 14.6 million occurrences annually!

[Help improve this site](#)

Annual Mean Thunderstorm Days (1993-2018)



Annual number of thunderstorm days in the U.S. From: Koehler, Thomas L., 2019: Cloud-to-Ground Lightning Flash Density and Thunderstorm Day Distributions over the Contiguous United States Derived from NLDN Measurements: 1993-2018. Under review at Monthly Weather Review. Used by permission.

[Download Image](#)

The figure above shows the average number of thunderstorm days each year throughout the U.S. (The graphic defines a day of thunderstorms as two lightning flashes within a 10 nmi. (nautical mile) (18.5 km) radius). The most frequent occurrence is in the southeastern states, with Florida having the highest number of "thunder" days (80 to 105+ days per year). This is because warm, moist air from the Gulf of Mexico and Atlantic Ocean (which we will see later are necessary ingredients for thunderstorm development) is most readily available to fuel thunderstorm development in this part of the country.

Last updated April 14, 2023

[Cite this page](#)